PES UNIVERSITY

Department of Computer Science & Engineering

Digital Forensics and Incident Responses

UE23CS342BA6

Assignment 1 Submission

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Topic: Unit 1 - Activity 1

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Digital Forensics and Incident Responses

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TASKS:

1. Locate the text file Tempest.txt. Without making any changes, review the contents of this file.

Activity 1 > Tasks > Tempest.txt

```
1   Be not afeard; the isle is full of noises,
2   Sounds and sweet airs, that give delight and hurt not.
3   Sometimes a thousand twangling instruments
4   Will hum about mine ears, and sometime voices
5   That, if I then had waked after long sleep,
6   Will make me sleep again: and then, in dreaming,
7   The clouds methought would open and show riches
8   Ready to drop upon me that, when I waked,
9   I cried to dream again.
10
```

2. Install and run the program HashCalc.
Since Hashcalc doesn't run on mac, we can use the terminal utilities for the same.

3. Simply drag and drop the file Tempest.txt into the HashCalc window.
a. What is the MD5 Hash value of this file?
Ans) 827193b76accd08a2540a0b33dc48eb4

```
pranavhemanth@Pranavs-MacBook-Pro-M3 Tasks %md5 Tempest.txt
MD5 (Tempest.txt) = 827193b76accd08a2540a0b33dc48eb4
pranavhemanth@Pranavs-MacBook-Pro-M3 Tasks %
```

b. What is the SHA1 Hash value of this file?
Ans) 53ed0c019f2fb9575765b72ddacec6226464ef65

```
pranavhemanth@Pranavs-MacBook-Pro-M3 Tasks %shasum -a 1 Tempest.txt
53ed0c019f2fb9575765b72ddacec6226464ef65  Tempest.txt
pranavhemanth@Pranavs-MacBook-Pro-M3 Tasks %
```

c. What is the SHA256 hash value of this file?
Ans) a5995872b7556a73e41a949373f5e944d6bb2e5d6ef924624d9ec3bf62666bac

```
pranavhemanth@Pranavs-MacBook-Pro-M3 Tasks %shasum -a 256 Tempest.txt
a5995872b7556a73e41a949373f5e944d6bb2e5d6ef924624d9ec3bf62666bac  Tempest.txt
pranavhemanth@Pranavs-MacBook-Pro-M3 Tasks %
```

4. Without changing any of the data in the text file, rename the file Bard.txt.

```
pranavhemanth@Pranavs-MacBook-Pro-M3 Tasks %cp Tempest.txt Bard.txt
pranavhemanth@Pranavs-MacBook-Pro-M3 Tasks %ls
~$ercise 1 - Hashing.docx       Collision.zip                FileSamples                 Tempest.txt                 USB1
Activity 1.pdf                  Exercise 1 - Hashing.docx    Image Files                 Tempest.zip                 USB1.zip
Bard.txt                        Files for hashing            README.md                   Tools
Collision                       Files for hashing.zip        Tasks                       Tools.zip
pranavhemanth@Pranavs-MacBook-Pro-M3 Tasks %
```

5. Using HashCalc recalculate the hash values and record them below:

a. MD5 hash value

Ans) 827193b76accd08a2540a0b33dc48eb4

```
pranavhemanth@Pranavs-MacBook-Pro-M3 Tasks %md5 Bard.txt
MD5 (Bard.txt) = 827193b76accd08a2540a0b33dc48eb4
pranavhemanth@Pranavs-MacBook-Pro-M3 Tasks %
```

b. SHA1 hash value

Ans) 53ed0c019f2fb9575765b72ddacec6226464ef65

```
pranavhemanth@Pranavs-MacBook-Pro-M3 Tasks %shasum -a 1 Bard.txt
53ed0c019f2fb9575765b72ddacec6226464ef65  Bard.txt
pranavhemanth@Pranavs-MacBook-Pro-M3 Tasks %
```

c. SHA256 hash value

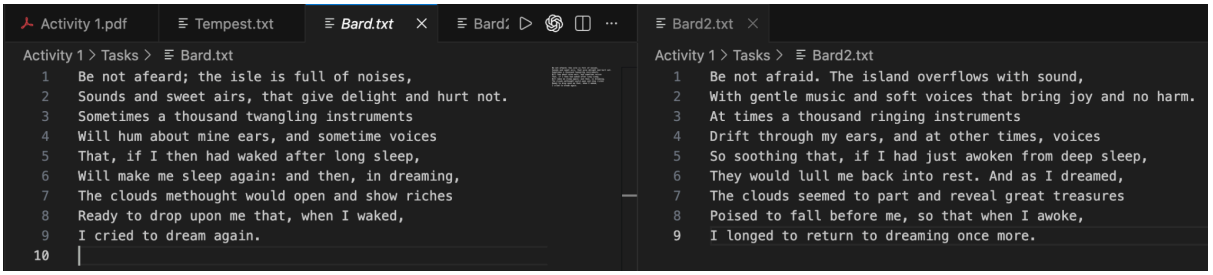
Ans) a5995872b7556a73e41a949373f5e944d6bb2e5d6ef924624d9ec3bf62666bac

```
pranavhemanth@Pranavs-MacBook-Pro-M3 Tasks %shasum -a 256 Bard.txt
a5995872b7556a73e41a949373f5e944d6bb2e5d6ef924624d9ec3bf62666bac  Bard.txt
pranavhemanth@Pranavs-MacBook-Pro-M3 Tasks %
```

6. What effect does a change of the file name have on the hash value?

Answer: The hash value does not change because the hashing algorithm operates on the file's contents, not its name. Since the data inside the file remains unchanged, the resulting hash stays the same.

7. Open the text file and do some changes. Save the file.



Bard2.txt is the changed file

Bard.txt

Activity 1 > Tasks > Bard.txt

```
1  Be not afeard; the isle is full of noises,
2  Sounds and sweet airs, that give delight and hurt not.
3  Sometimes a thousand twangling instruments
4  Will hum about mine ears, and sometime voices
5  That, if I then had waked after long sleep,
6  Will make me sleep again: and then, in dreaming,
7  The clouds methought would open and show riches
8  Ready to drop upon me that, when I waked,
9  I cried to dream again.
10
```

Bard2.txt

Activity 1 > Tasks > Bard2.txt

```
1  Be not afraid. The island overflows with sound,
2  With gentle music and soft voices that bring joy and no harm.
3  At times a thousand ringing instruments
4  Drift through my ears, and at other times, voices
5  So soothing that, if I had just awoken from deep sleep,
6  They would lull me back into rest. And as I dreamed,
7  The clouds seemed to part and reveal great treasures
8  Poised to fall before me, so that when I awoke,
9  I longed to return to dreaming once more.
```

a. What is the MD5 Hash value of this file?
Ans) 9e20d6d66af08068811e13e73c5962e7

```
pranavhemanth@Pranavs-MacBook-Pro-M3 Tasks %md5 Bard2.txt
MD5 (Bard2.txt) = 9e20d6d66af08068811e13e73c5962e7
pranavhemanth@Pranavs-MacBook-Pro-M3 Tasks %
```

b. What is the SHA1 Hash value of this file?
Ans) 463f84a01a0fa1df9226ee45564c510138f56be5

```
pranavhemanth@Pranavs-MacBook-Pro-M3 Tasks %shasum -a 1 Bard2.txt
463f84a01a0fa1df9226ee45564c510138f56be5  Bard2.txt
pranavhemanth@Pranavs-MacBook-Pro-M3 Tasks %
```

c. What is the SHA256 hash value of this file?
Ans) 0e881492798029e2edeb9a9d247645539d252e7abdfdd8a499130fb9a8eb94ca

```
pranavhemanth@Pranavs-MacBook-Pro-M3 Tasks %shasum -a 256 Bard2.txt
0e881492798029e2edeb9a9d247645539d252e7abdfdd8a499130fb9a8eb94ca  Bard2.txt
pranavhemanth@Pranavs-MacBook-Pro-M3 Tasks %
```

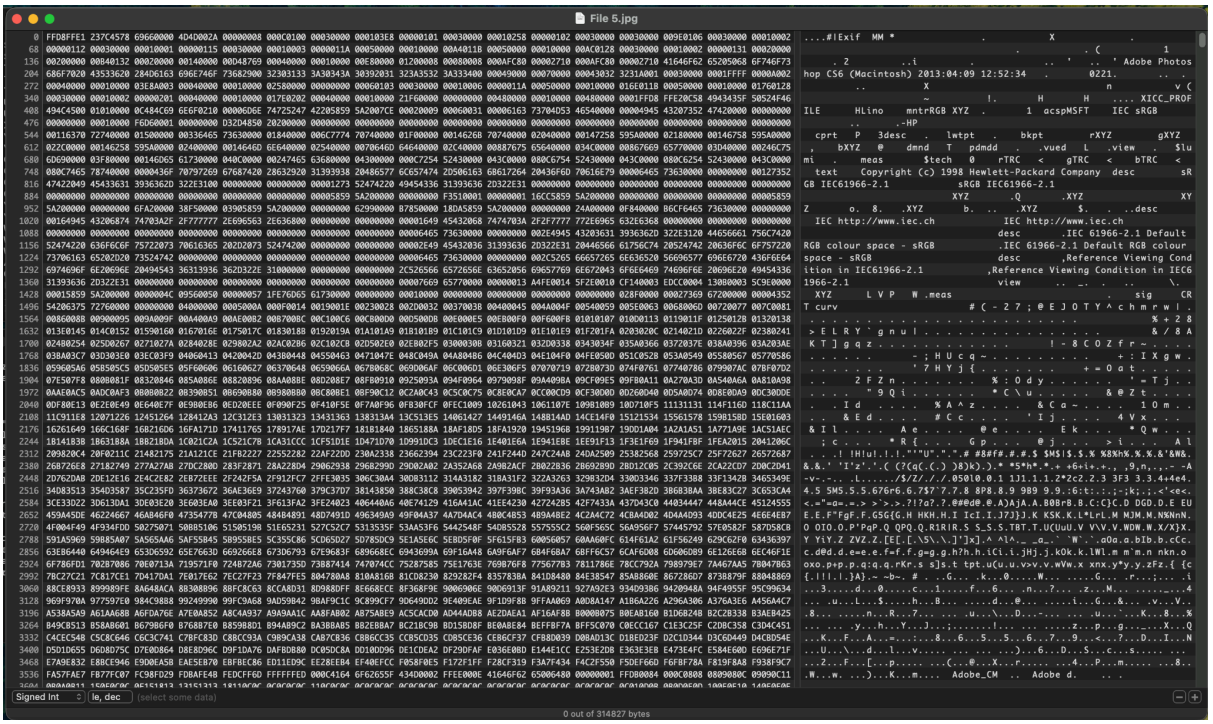
8. Calculate the Hash value of the files in the Files for hashing folder and record them below:

```
pranavhemanth@Pranavs-MacBook-Pro-M3 Tasks %cd Files\ for\ hashing
pranavhemanth@Pranavs-MacBook-Pro-M3 Files for hashing %md5 File\ 1.jpg
MD5 (File 1.jpg) = 2e0bf5e90ea329adc1e4427a30010ee3
pranavhemanth@Pranavs-MacBook-Pro-M3 Files for hashing %md5 File\ 2.jpg
MD5 (File 2.jpg) = 32216ccffb6f3f2184a54ce11f537a11
pranavhemanth@Pranavs-MacBook-Pro-M3 Files for hashing %md5 File\ 3.jpg
MD5 (File 3.jpg) = d3bb46fde0d7669b71a485b0473e226c
pranavhemanth@Pranavs-MacBook-Pro-M3 Files for hashing %md5 File\ 4.jpg
MD5 (File 4.jpg) = 55e8744a4e9e9ec4c72a9989e32e3e0a
pranavhemanth@Pranavs-MacBook-Pro-M3 Files for hashing %md5 File\ 5.jpg
MD5 (File 5.jpg) = 0f1147ee86747286ee1df5a5f6fb2e19
pranavhemanth@Pranavs-MacBook-Pro-M3 Files for hashing %
```

```
pranavhemanth@Pranavs-MacBook-Pro-M3 Files for hashing %shasum -a 1 File\ 1.jpg
8ea86862f53a873f8f3cb2449e6d7cd4af29a85a  File 1.jpg
pranavhemanth@Pranavs-MacBook-Pro-M3 Files for hashing %shasum -a 1 File\ 2.jpg
823ca7e5a6bbdfb9ec9a7bbb2412f83c1e590293  File 2.jpg
pranavhemanth@Pranavs-MacBook-Pro-M3 Files for hashing %shasum -a 1 File\ 3.jpg
ec067b0391b769ba54d0c66e321520776fa89f47  File 3.jpg
pranavhemanth@Pranavs-MacBook-Pro-M3 Files for hashing %shasum -a 1 File\ 4.jpg
d674a23c2b807bb2cb1be878839e47c06ba67c27  File 4.jpg
pranavhemanth@Pranavs-MacBook-Pro-M3 Files for hashing %shasum -a 1 File\ 5.jpg
b8da917c2468c98d77d9cd4d60a104dbd1af2196  File 5.jpg
pranavhemanth@Pranavs-MacBook-Pro-M3 Files for hashing %
```

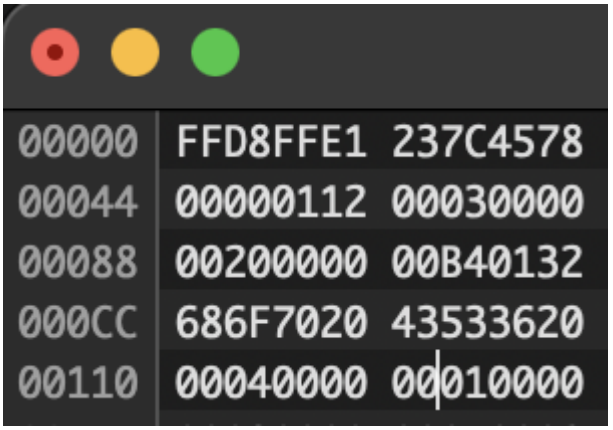
File Name	MD5	SHA1
File 1.jpg	2e0bf5e90ea329adc1e4427a30010ee3	8ea86862f53a873f8f3cb2449e6d7cd4af29a85a
File 2.jpg	32216ccffb6f3f2184a54ce11f537a11	823ca7e5a6bbdfb9ec9a7bbb2412f83c1e590293
File 3.jpg	d3bb46fde0d7669b71a485b0473e226c	ec067b0391b769ba54d0c66e321520776fa89f47
File 4.jpg	55e8744a4e9e9ec4c72a9989e32e3e0a	d674a23c2b807bb2cb1be878839e47c06ba67c27
File 5.jpg	0f1147ee86747286ee1df5a5f6fb2e19	b8da917c2468c98d77d9cd4d60a104dbd1af2196

9. Install and run the program HxD and open the file File5.jpg
We can use Hex Fiend in mac instead of HxD as a hex editor

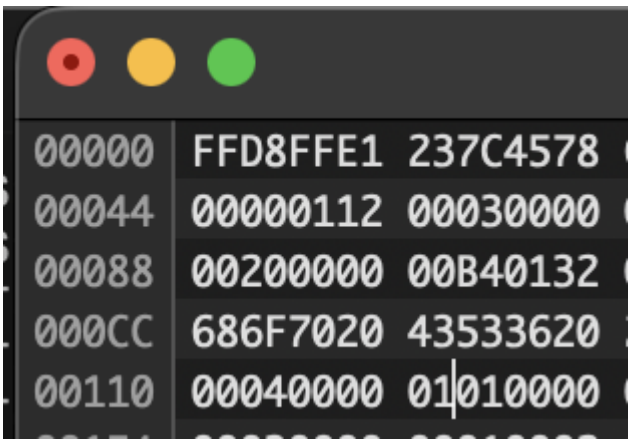


10. Navigate to file offset (h)114 and change the value from 00 to 01 (you are changing just one bit of data – from 00000000 to 00000001)

Before:

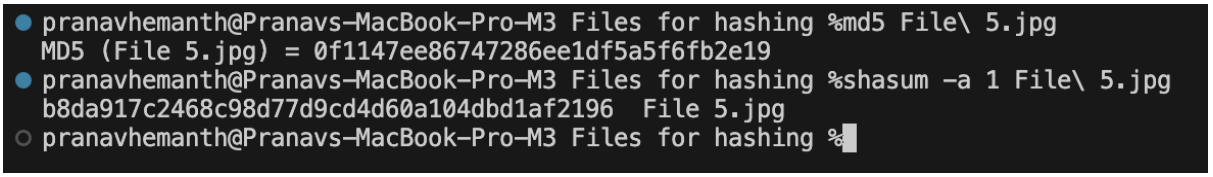


After:



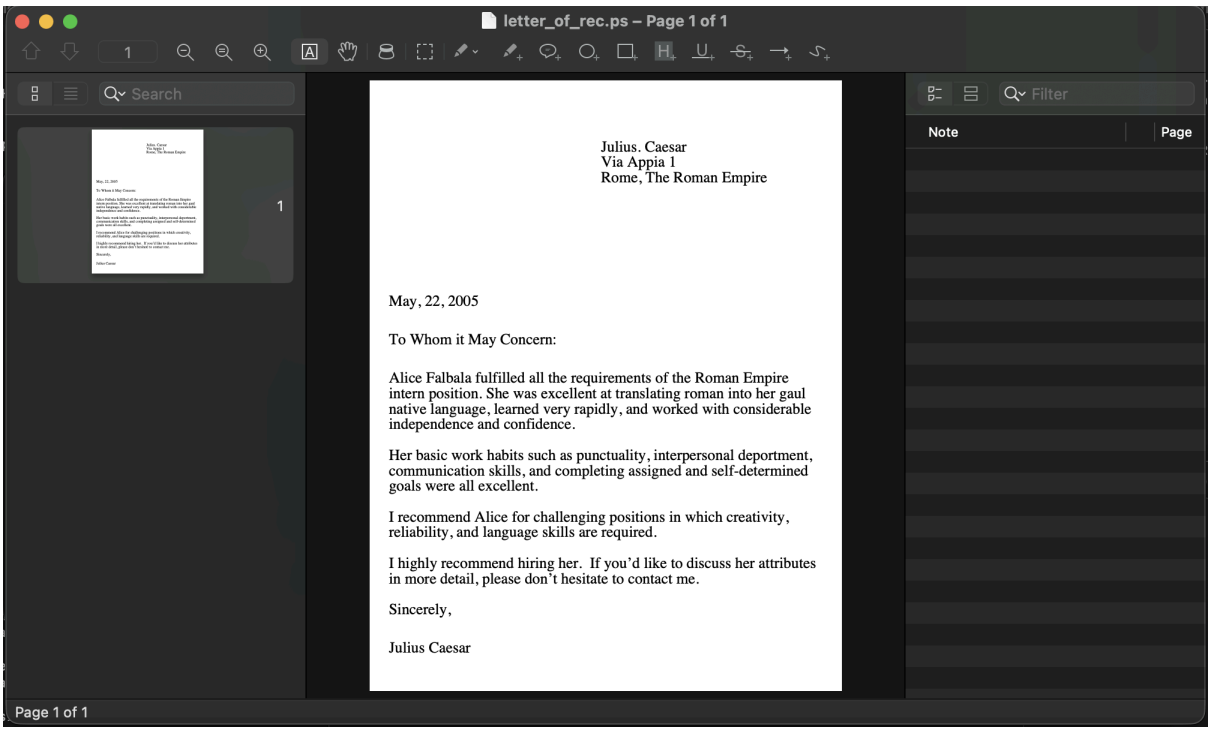
11. Save the file as New File 5.jpg

12. Calculate the hash values of this new file.
a. MD5: 06ca4f972ff4aa3000565a864811849d
a. MD5: 0f1147ee86747286ee1df5a5f6fb2e19
b. SHA1: 475329763117c5a57e1bee7968e9caa473692a0d
b. SHA1: b8da917c2468c98d77d9cd4d60a104dbd1af2196

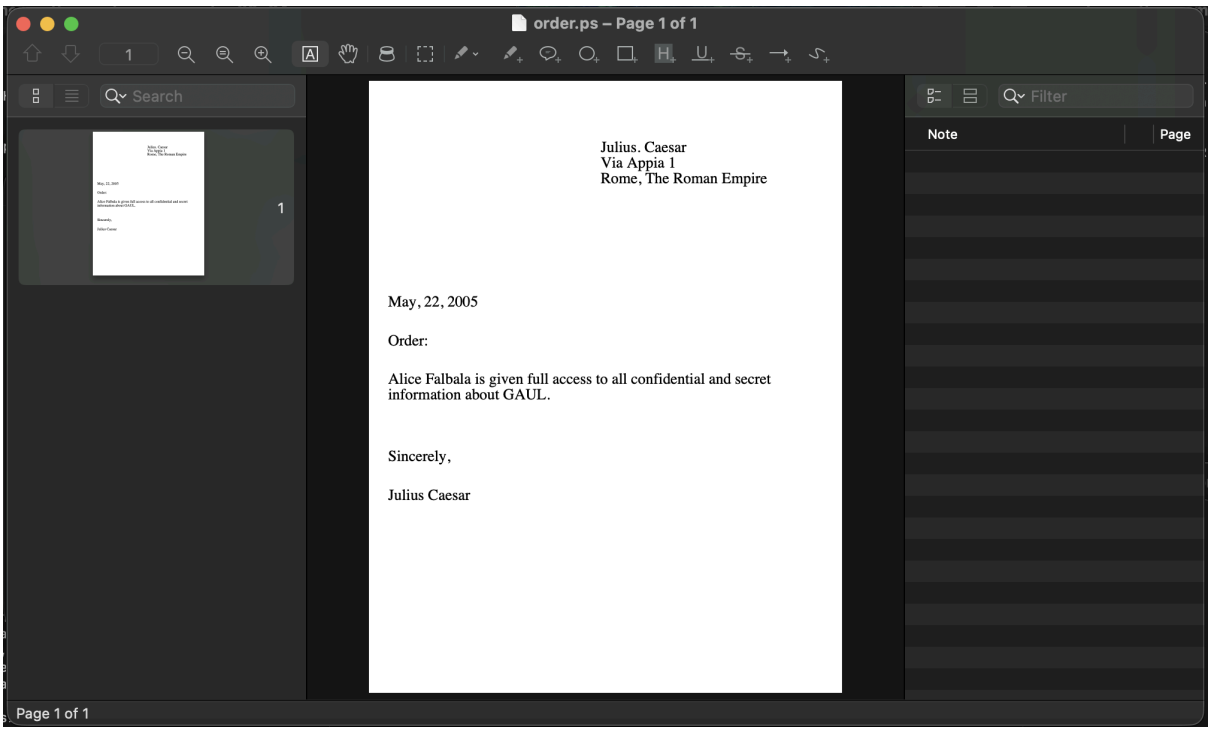


13. Using the program EPSViewer open the files letter_of_rec.ps and the file order.ps which are in the collision folder.
We can use skim on macos instead of EPS viewer for .ps files

letter_of_rec.ps:



order.ps:



14. Briefly describe the content of these files

a. letter_of_rec.ps

Answer: It is a formal letter of recommendation written by Julius Caesar for Alice Falbala, praising her performance as an intern of the Roman Empire, her language skills, work ethic, and professionalism, and strongly recommending her for future positions.

b. order.ps

Answer: It is an official order from Julius Caesar granting Alice Falbala full access to confidential and secret information about Gaul.

So we can clearly see that the files are different. We can do a simple comparison of both the files as shown below:

```
pranavhemanth@Pranavs-MacBook-Pro-M3 Tasks %cd Collision
pranavhemanth@Pranavs-MacBook-Pro-M3 Collision %ls
letter_of_rec.ps        order.ps
pranavhemanth@Pranavs-MacBook-Pro-M3 Collision %cmp letter_of_rec.ps order.ps
letter_of_rec.ps order.ps differ: char 84, line 3
pranavhemanth@Pranavs-MacBook-Pro-M3 Collision %shasum -a 256 letter_of_rec.ps
de4e4c6e2b94e95a3c5bd72a9a6af29bc5f83bf759325d9921943a6fc08ea245  letter_of_rec.ps
pranavhemanth@Pranavs-MacBook-Pro-M3 Collision %shasum -a 256 order.ps
077046dd66015e05c3e03a43a6e4de129038e0701de5a4103fc7ed91c3782d06  order.ps
pranavhemanth@Pranavs-MacBook-Pro-M3 Collision %
```

15. Calculate the MD5 hash value of these two files.

```
pranavhemanth@Pranavs-MacBook-Pro-M3 Collision %md5 letter_of_rec.ps
MD5 (letter_of_rec.ps) = a25f7f0b29ee0b3968c860738533a4b9
pranavhemanth@Pranavs-MacBook-Pro-M3 Collision %md5 order.ps
MD5 (order.ps) = a25f7f0b29ee0b3968c860738533a4b9
pranavhemanth@Pranavs-MacBook-Pro-M3 Collision %
```

File Name	MD5
letter_of_rec.ps	a25f7f0b29ee0b3968c860738533a4b9
order.ps	a25f7f0b29ee0b3968c860738533a4b9

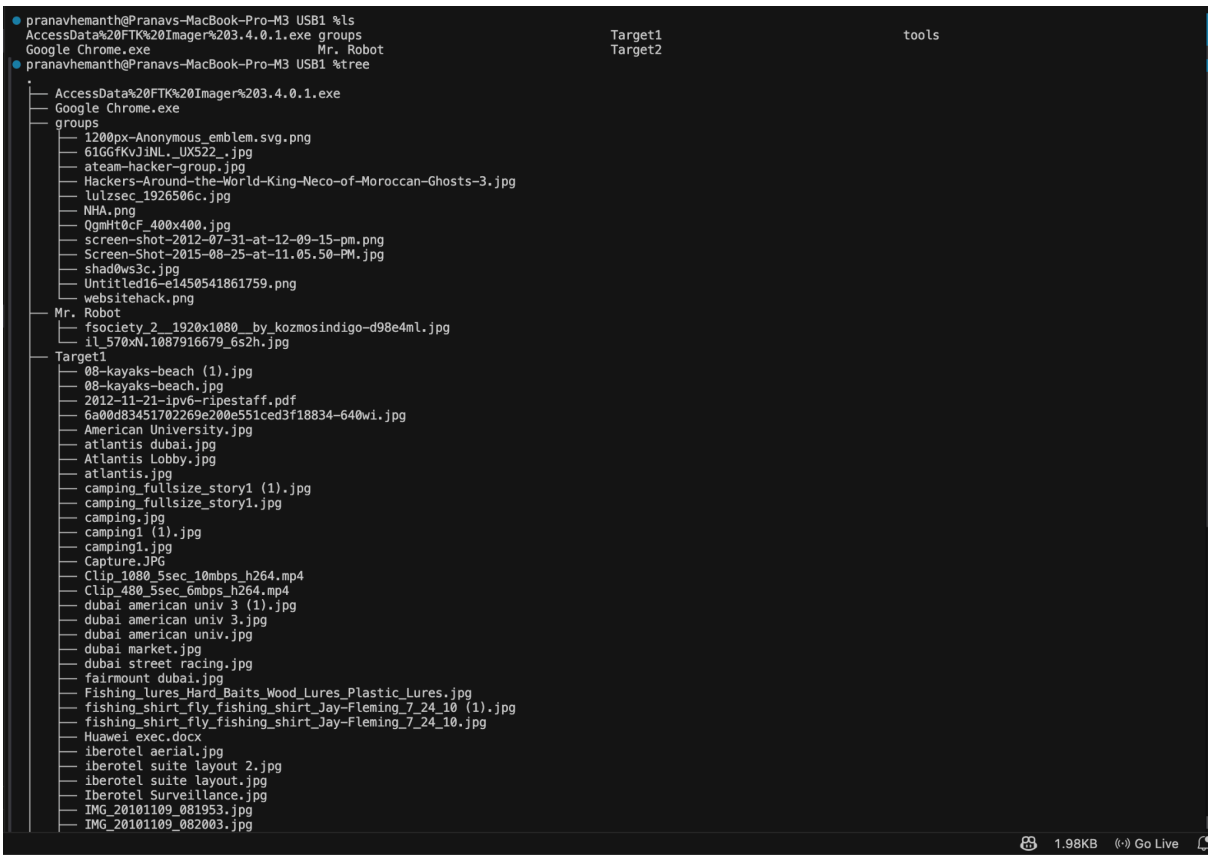
We are able to observe a collision in the md5 hashing of the two files above

16. These two files seem to have different content but the same MD5 hash value. What is this called?
Answer: A hash collision (specifically an MD5 collision)

17. How might you rectify this issue?
Answer: This problem can be addressed by replacing MD5 with a collision resistant cryptographic hash function such as SHA-256. In addition, using multiple hashes or digital signatures for integrity checks helps prevent attackers from taking advantage of hash collisions.

18. The folder USB1 contains files recovered from a suspect's thumb drive. It is believed that there may be files of value on this thumb drive. Previous examinations have identified files of value with the following MD5 hash values: 4327397f1854d452a8a4c8dc4776738
3025d7ea8904fdaa00e795286bcbe651

19. Examine the files in the USB1 folder and identify any files that match these hash values.



There are clearly a lot of files

We can speedup the process of checking by just hashing all files and crosschecking

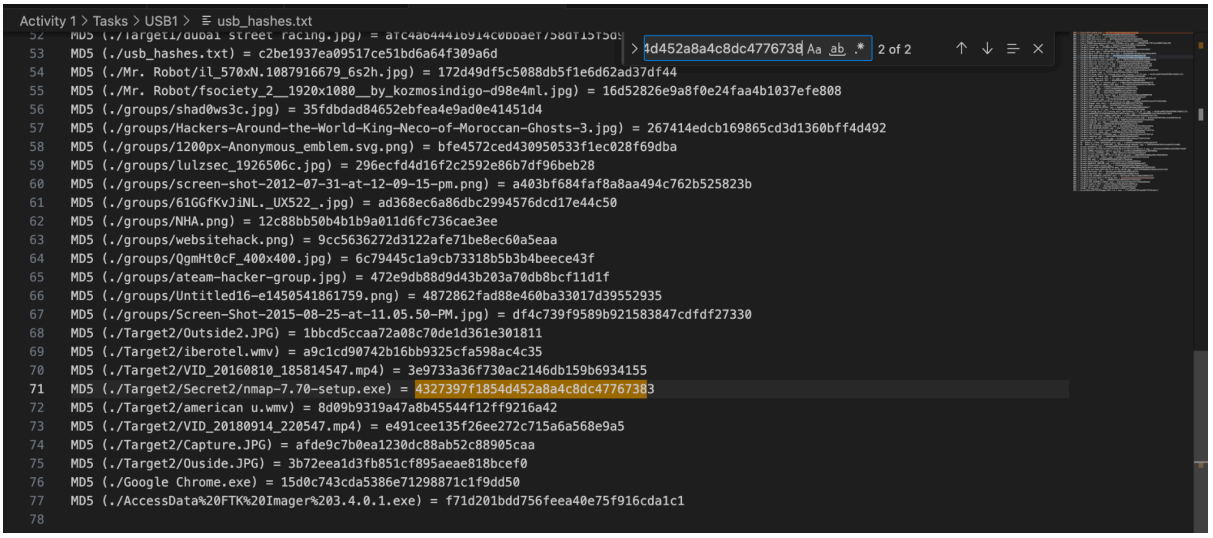
```
pranavhemanth@Pranavs-MacBook-Pro-M3 USB1 %md5 *
MD5 (AccessData%20FTK%20Imager%203.4.0.1.exe) = f71d201bdd756feea40e75f916cda1c1
MD5 (Google Chrome.exe) = 15d0c743cda5386e71298871c1f9dd50
md5: groups: Is a directory
md5: Mr. Robot: Is a directory
md5: Target1: Is a directory
md5: Target2: Is a directory
md5: tools: Is a directory
pranavhemanth@Pranavs-MacBook-Pro-M3 USB1 %
```

No match here.. need to check inside the directories

We can recursively hash and pipe the results to a file and then compare:

```
pranavhemanth@Pranavs-MacBook-Pro-M3 USB1 %md5 *
pranavhemanth@Pranavs-MacBook-Pro-M3 USB1 %find . -type f -exec md5 {} \; > usb_hashes.txt
pranavhemanth@Pranavs-MacBook-Pro-M3 USB1 %ls
AccessData%20FTK%20Imager%203.4.0.1.exe groups                                  Target1                                 tools
Google Chrome.exe                       Mr. Robot                               Target2                                 usb_hashes.txt
pranavhemanth@Pranavs-MacBook-Pro-M3 USB1 %cat usb_hashes.txt
MD5 (./tools/Office2010.exe) = 4327397f1854d452a8a4c8dc47767383
MD5 (./tools/images.jfif) = 591f07a4de8394d02c836225537a6e47
MD5 (./tools/download.jfif) = a4f73e8bb339cb66f5d286428433a0c7
MD5 (./tools/Job1/Solitaire.exe) = 86e7586e4e45444f23ef2b71e2a93bfb
MD5 (./tools/QuickTime Installer.exe) = 15d0c743cda5386e71298871c1f9dd50
MD5 (./tools/120302-pwnieexpress-017edit-4f524be-intro.jpg) = 1080297f99c7707c5efd5087f8d2c290
MD5 (./Target1/fairmount dubai.jpg) = 942ebf7ffe7a9e28338de17d3bea95bb
MD5 (./Target1/Thumbs.db) = 50f98bd74a4f04f7731bbacb7d388f95
MD5 (./Target1/dubai american univ.jpg) = 4057929ecb19b1cbb94a3a22eba36db3
MD5 (./Target1/street.jpg) = b407ddc7f2993b33c2f4b75663dda738
MD5 (./Target1/Maid with the Flaxen Hair.mp3) = 5bacbdba9af42150c27b1a182ba169f8
MD5 (./Target1/08-kayaks-beach.jpg) = 74cce51e77a7c08bc5d74db6d2453c96
MD5 (./Target1/08-kayaks-beach (1).jpg) = 74cce51e77a7c08bc5d74db6d2453c96
MD5 (./Target1/2012-11-21-ipv6-ripestaff.pdf) = f4df5312314dd1e2830c2c8b3e3d429b
MD5 (./Target1/American University.jpg) = e3095d33741743fe62e3ddfaafa6866b
MD5 (./Target1/Iberotel Surveillance.jpg) = fc0a4a29ea8102941bd5f0c3cc35f922
MD5 (./Target1/dubai american univ 3.jpg) = bd5fb86115f40c4a37b6c3587d39c397
MD5 (./Target1/islamabad-arrival2.jpg) = c1c10bbf372c38f7a5130475b4e96be0
MD5 (./Target1/camping_fullsize_story1 (1).jpg) = 5b0df61cd81ab44e4d2ecef5fb597b54
MD5 (./Target1/atlantis.jpg) = 74f13f7cd5e176c932acf288bc224541
MD5 (./Target1/fishing_shirt_fly_fishing_shirt_Jay-Fleming_7_24_10.jpg) = 5bc02ce6982346a30f090c25b661c17d
MD5 (./Target1/atlantis dubai.jpg) = 04836e8891b71806583ba4c57e43c60e
MD5 (./Target1/dubai american univ 3 (1).jpg) = bd5fb86115f40c4a37b6c3587d39c397
MD5 (./Target1/Walden_Kayaks2.JPG) = 3d5a983b88dd42abb2732d6b3d823638
MD5 (./Target1/PowerPoint is Awesome.pptx) = efdbbc36b4b1a49568dc094a99ebef76
MD5 (./Target1/Clip_480_5sec_6mbps_h264.mp4) = 77f7344d77e4fee542c92741756670cc
MD5 (./Target1/kayak1.jpg) = fc6aeb127ccf63ba84901c2daa027d3a
MD5 (./Target1/camping.jpg) = f7ad11c8b00d99d2a0c32e96a3775746
MD5 (./Target1/Kayak2.jpg) = 14671de93ac33069bac6f7adf6877853
MD5 (./Target1/Kayaks.jpg) = 70a47f3bae749df5b7c99d41b8ec41a3
MD5 (./Target1/iberotel suite layout.jpg) = 0c53d42fa31baf3c689eada908089706
MD5 (./Target1/iberotel aerial.jpg) = 327825a2894b2e81dc39d4125ab6ae29
MD5 (./Target1/sea-kayaks.jpg) = cb7f6c9913a2b0cad8a1c619097e4d09
MD5 (./Target1/Islamabad-MarriottHotel-Exterior-02.jpg) = c9488dcc8ed484314eaf5472a95ee0ba
MD5 (./Target1/Huawei exec.docx) = b4cebdfd5a9723e70dd1526377b675de
MD5 (./Target1/dubai market.jpg) = 646259d5f67ddbe59fb27f2026cb8374
MD5 (./Target1/IMG_20101109_081953.jpg) = 68a65f342b6c09b5a8e7b80824503b77
MD5 (./Target1/sea-kayaks (1).jpg) = cb7f6c9913a2b0cad8a1c619097e4d09
MD5 (./Target1/fishing_shirt_fly_fishing_shirt_Jay-Fleming_7_24_10 (1).jpg) = 5bc02ce6982346a30f090c25b661c17d
MD5 (./Target1/6a00d83451702269e200e551ced3f18834-640wi.jpg) = 77b1400d999b713dc78bf9c455773620
MD5 (./Target1/Clip_1080_5sec_10mbps_h264.mp4) = 5c3f6fcd49941b6c304ebe6c76effc4a
MD5 (./Target1/Fishing_lures_Hard_Baits_Wood_Lures_Plastic_Lures.jpg) = 5fe3e219ef0a578eeca1bad50155acd2
MD5 (./Target1/camping_fullsize_story1.jpg) = 5b0df61cd81ab44e4d2ecef5fb597b54
MD5 (./Target1/camping1 (1).jpg) = 6180f436e7031260398f99de46c84394
MD5 (./Target1/Capture.JPG) = 86905b3481391893e08b7df08a1c97c8
MD5 (./Target1/IMG_20150709_132031655.jpg) = 03ce4862e0f030cd37412204fee33b1b
MD5 (./Target1/IMG_20101109_082003.jpg) = b3af191bffbf75eec38fd34ecf74af42
MD5 (./Target1/iberotel suite layout 2.jpg) = 33186f75b9c794489babda577e0fbfee
MD5 (./Target1/Atlantis Lobby.jpg) = 251d740b7be72fb11709722d63793520
MD5 (./Target1/IMG_20160403_124618460.jpg) = 90f9543df6dbe5e367d9e2a0374b3fa6
```

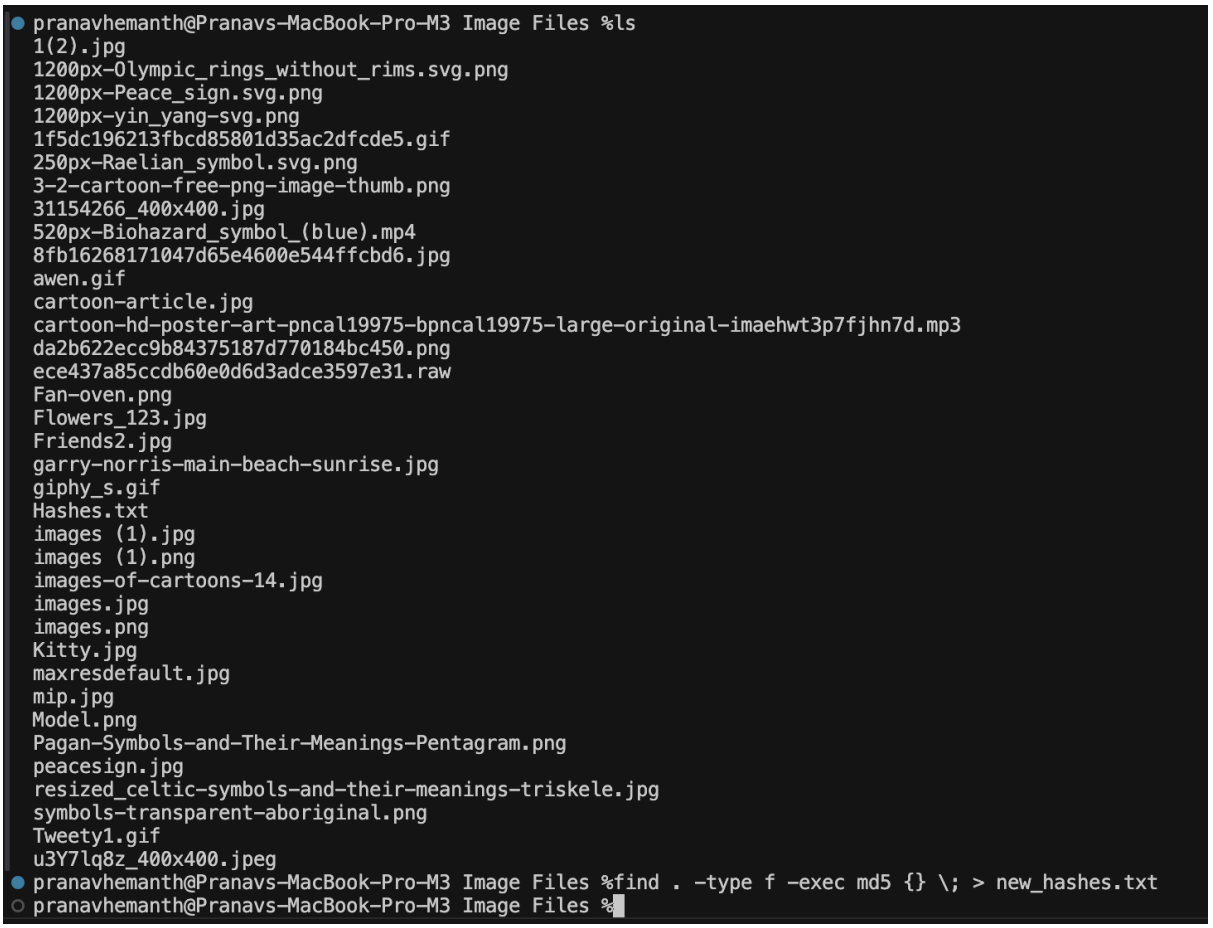
First match - 4327397f1854d452a8a4c8dc4776738



File match = ./Target2/Secret2/nmap-7.70-setup.exe)

Second match - 3025d7ea8904fdaa00e795286bcbe651
Not found

20. Generate the MD5 hash value of selected files (files in Image Files Folder) using MD5 Calculator Compare the generated hash values of files with their pre-existing hash values (look into hashes.txt) to determine the integrity of files



Activity 1 > Tasks > Image Files > Hashes.txt

```
1   cartoon-article.jpg: 910ff95c4076b4555e7241f7a7411ebb
2   Fan-oven.png: 3759b97fbd029a7e74a492fd09bc0a3d
3   Flowers_123.jpg: eadd66a945bebf7eb01b2a75a1748f99
4   Kitty.jpg: bf0c7d9a99c575941221ef0c2ff604fb
5   Model.png: 209c1f5a72f2181b5a670e45cbd550f6
6   peacesign.jpg: 9cfda073866cd6f813815ef6b97e3c32
7   images.jpg: 52b29a36da797ff3c4c9fed18140176b
8   awen.gif: 169f012153f9e70767bbbeddae427fbf
9   Friends2.jpg: 7c6f8ea5361ae7719d5d3caf60bf60b5
10  Tweety1.gif: 3ef2b01136671ae97d3ccf4d22cf58c1
11  images-of-cartoons-14.jpg: e0413409680f4e48ea7b68e725c36a4a
12  maxresdefault.jpg: 72fdbe03e810f94e7f066f986181d72f
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
pranavhemanth@Pranavs-MacBook-Pro-M3 Image Files %md5 cartoon-article.jpg
MD5 (cartoon-article.jpg) = 910ff95c4076b4555e7241f7a7411ebb
pranavhemanth@Pranavs-MacBook-Pro-M3 Image Files %md5 Fan-oven.png
MD5 (Fan-oven.png) = 3759b97fbd029a7e74a492fd09bc0a3d
pranavhemanth@Pranavs-MacBook-Pro-M3 Image Files %md5 Flowers_123.jpg
MD5 (Flowers_123.jpg) = eadd66a945bebf7eb01b2a75a1748f99
pranavhemanth@Pranavs-MacBook-Pro-M3 Image Files %md5 Kitty.jpg
MD5 (Kitty.jpg) = bf0c7d9a99c575941221ef0c2ff604fb
pranavhemanth@Pranavs-MacBook-Pro-M3 Image Files %md5 Model.png
MD5 (Model.png) = d62fb1819eeaf2315fdc0ade6e10ed25
pranavhemanth@Pranavs-MacBook-Pro-M3 Image Files %md5 peacesign.jpg
MD5 (peacesign.jpg) = 9cfda073866cd6f813815ef6b97e3c32
pranavhemanth@Pranavs-MacBook-Pro-M3 Image Files %md5 images.jpg
MD5 (images.jpg) = 52b29a36da797ff3c4c9fed18140176b
pranavhemanth@Pranavs-MacBook-Pro-M3 Image Files %md5 awen.gif
MD5 (awen.gif) = 169f012153f9e70767bbbeddae427fbf
pranavhemanth@Pranavs-MacBook-Pro-M3 Image Files %md5 Friends2.jpg
MD5 (Friends2.jpg) = 7c6f8ea5361ae7719d5d3caf60bf60b5
pranavhemanth@Pranavs-MacBook-Pro-M3 Image Files %md5 Tweety1.gif
MD5 (Tweety1.gif) = 3ef2b01136671ae97d3ccf4d22cf58c1
pranavhemanth@Pranavs-MacBook-Pro-M3 Image Files %md5 images-of-cartoons-14.jpg
MD5 (images-of-cartoons-14.jpg) = e0413409680f4e48ea7b68e725c36a4a
pranavhemanth@Pranavs-MacBook-Pro-M3 Image Files %md5 maxresdefault.jpg
MD5 (maxresdefault.jpg) = 72fdbe03e810f94e7f066f986181d72f
pranavhemanth@Pranavs-MacBook-Pro-M3 Image Files %
```

As we can observe the only difference is the file - Model.png
old hash - 209c1f5a72f2181b5a670e45cbd550f6
new hash - d62fb1819eeaf2315fdc0ade6e10ed25

This signals some loss of integrity. The file may be tampered with

21. Using HxD verify the filetypes given in the FileSamples Folder.
Reference table below:

FILE SIGNATURES TABLE

File Extension	Hex Signature	ASCII Signature	Trailer	File Description
DOC, DOT, PPS, PPT, XLA, XLS, WIZ	D0 CF 11 E0 A1 B1 1A E1	ÐÏ.à¡±.á		An Object Linking and Embedding (OLE) Compound File (CF) (i.e., OLECF) file format, known as Compound Binary File format by Microsoft, used by Microsoft Office 97-2003 applications (Word, Powerpoint, Excel, Wizard). [See also Word, Powerpoint, and Excel "subheaders" at byte offset 512. There appear to several subheader formats and a dearth of documentation. There have been reports that there are different subheaders for Windows and Mac versions of MS Office but I cannot confirm that.] [Note the similarity between D0 CF 11 E0 and the word "docfile"!] AC_ CaseWare Working Papers compressed client file ADP Access project file APR Lotus/IBM Approach 97 file DB MSWorks database file MSC Microsoft Common Console Document MSI Microsoft Installer package MTW Minitab data file OPT Developer Studio File Workspace Options file PUB MS Publisher file QBM QuickBooks Portable Company File RVT Revit Project file SOU Visual Studio Solution User Options file SPO SPSS output file VSD Visio file WPS MSWorks text document
DOC	[512 byte offset] EC A5 C1 00	[512 byte offset] ì¥Á.		Word document subheader (MS Office)
XLS	[512 byte offset] FD FF FF FF nn 00 or [512 byte offset] FD FF FF FF nn 02 or [512 byte offset] 09 08 10 00 00 06 05 00	[512 byte offset] ýÿÿÿ.. or [512 byte offset] ýÿÿÿ.. or [512 byte offset] ……		Excel spreadsheet subheader (MS Office)
PPT	[512 byte offset] A0 46 1D F0 or [512 byte offset] 00 6E 1E F0 or [512 byte offset] 0F 00 E8 03 or [512 byte offset] FD FF FF FF nn nn 00 00	[512 byte offset] F.ð or [512 byte offset] .n.ð or [512 byte offset] ..è. or [512 byte offset] ýÿÿÿ....		PowerPoint presentation subheader (MS Office)
DOCX, PPTX, XLSX	50 4B 03 04 14 00 06 00	PK.....	Look for 50 4B 05 06 (PK..) followed by 18 additional bytes at the end of the file.	Microsoft Office Open XML Format (OOXML) Document NOTE: There is no subheader for MS OOXML files as there is with DOC, PPT, and XLS files. To better understand the format of these files, rename any OOXML file to have a .ZIP extension and then unZIP the file; look at the resultant file named [Content_Types].xml to see the content types. In particular, look for the <Override PartName= tag, where you will find word, ppt, or xl, respectively.
JPG	FF D8	ÿØ	FF D9	Digital camera JPG
PNG	89 50 4E 47 0D 0A 1A 0A	‰PNG....	49 45 4E 44 AE 42 60 82 (IEND®B`,...)	Portable Network Graphics file
PDF	25 50 44 46	%PDF	0A 25 25 45 4F 46 (.%%EOF) 0A 25 25 45 4F 46 0A (.%%EOF.) 0D 0A 25 25 45 4F 46 0D 0A (..%%EOF..) 0D 25 25 45 4F 46 0D (.%%EOF.) NOTE: There may be multiple end-of-file marks within the file. When carving, be sure to get the last one.	Adobe Portable Document Format and Forms Document file
ZIP	50 4B 03 04		Trailer: filename 50 4B 17 characters 00 00 00 Trailer: (filename PK 17 characters ...)	KZIP archive file
PST	21 42 44 4E	!BDN		Microsoft Outlook Personal Folder File

1.ppt



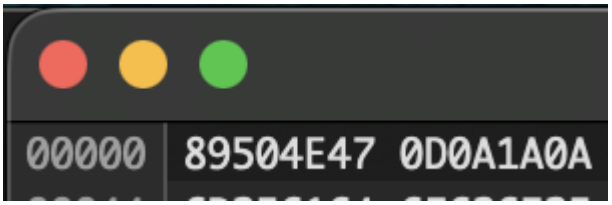
But this sequence is for a pdf file not a ppt. We now know that the file is actually a pdf not a ppt. We can confirm this with binwalk

```
pranavhemanth@Pranavs-MacBook-Pro-M3 FileSamples %ls
File Analysis                File Analysis.zip            FILE SIGNATURES TABLE.pdf
pranavhemanth@Pranavs-MacBook-Pro-M3 FileSamples %cd File\ Analysis
pranavhemanth@Pranavs-MacBook-Pro-M3 File Analysis %ls
1.ppt   2.image 3.html  4.txt   5.pdf   6.php   7.svg   8       9
pranavhemanth@Pranavs-MacBook-Pro-M3 File Analysis %binwalk 1.ppt

                        /Users/pranavhemanth/Code/Academics/DFIR-S6/Activity 1/Tasks/Tasks/FileSamples/File Analysis/1.ppt
--------------------------------------------------------------------------------
DECIMAL                         HEXADECIMAL                     DESCRIPTION
--------------------------------------------------------------------------------
0                               0x0                             PDF document, version 1.7
1861                            0x745                           JPEG image, total size: 24708 bytes
30265                           0x7639                          JPEG image, total size: 10487 bytes
43406                           0xA98E                          JPEG image, total size: 14334 bytes
61502                           0xF03E                          JPEG image, total size: 14107 bytes
--------------------------------------------------------------------------------

Analyzed 1 file for 85 file signatures (187 magic patterns) in 5.0 milliseconds
pranavhemanth@Pranavs-MacBook-Pro-M3 File Analysis %
```

2.image



This is as per the table and is a png file. Binwalk below

```
pranavhemanth@Pranavs-MacBook-Pro-M3 File Analysis %ls
1.ppt   2.image 3.html  4.txt   5.pdf   6.php   7.svg   8       9
pranavhemanth@Pranavs-MacBook-Pro-M3 File Analysis %binwalk 2.image

                        /Users/pranavhemanth/Code/Academics/DFIR-S6/Activity 1/Tasks/Tasks/FileSamples/File Analysis/2.image
--------------------------------------------------------------------------------
DECIMAL                         HEXADECIMAL                     DESCRIPTION
--------------------------------------------------------------------------------
0                               0x0                             PNG image, total size: 215538 bytes
--------------------------------------------------------------------------------

Analyzed 1 file for 85 file signatures (187 magic patterns) in 10.0 milliseconds
pranavhemanth@Pranavs-MacBook-Pro-M3 File Analysis %
```

3.html

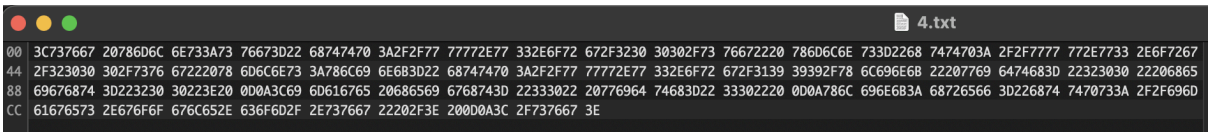
```
3.html
000 3C68746D 6C3E0D0A 0D0A2020 3C212D2D 20435352 4620506F 43202D20 67656E65 72617465 64206279 20427572 70205375 69746520 50726F66 65737369 6F6E616C 202D2D3E
044 0D0A0D0A 20203C62 6F64793E 0D0A0D0A 20203C73 63726970 743E6869 73746F72 792E7075 73685374 61746528 27272C20 27272C20 272F2729 3C2F7363 72697074 3E0D0A0D
088 0A202020 203C666F 726D2061 6374696F 6E3D2268 74747073 3A2F2F77 77772E67 616D6269 61656D61 726B6574 2E636F6D 2F757365 7222206D 6574686F 643D2250 4F535422
0CC 3E0D0A0D 0A202020 2020203C 696E7075 74207479 70653D22 68696464 656E2220 6E616D65 3D226373 72662623 39353B74 65737426 2339353B 6E616D65 22207661 6C75653D
110 22613437 63386139 34393436 65376139 30653130 31646438 36383162 32646164 3122202F 3E0D0A0D 0A202020 2020203C 696E7075 74207479 70653D22 68696464 656E2220
154 6E616D65 3D22656D 61696C22 2076616C 75653D22 74657374 26233634 3B676D61 696C2623 34363B63 6F6D2220 2F3E0D0A 0D0A2020 20202020 3C696E70 75742074 7970653D
198 22686964 64656E22 206E616D 653D2270 61737377 6F726422 2076616C 75653D22 26233634 3B262336 343B6D4D 31323334 35363738 22202F3E 0D0A0D0A 20202020 20203C69
1DC 6E707574 20747970 653D2268 69646465 6E22206E 616D653D 226C6F67 696E2220 76616C75 653D2222 202F3E0D 0A0D0A20 20202020 203C696E 70757420 74797065 3D227375
220 626D6974 22207661 6C75653D 22537562 6D697420 72657175 65737422 202F3E0D 0A0D0A20 2020203C 2F666F72 6D3E0D0A 0D0A2020 3C2F626F 64793E0D 0A0D0A3C 2F68746D
264 6C3E0D0A 0D0A
```

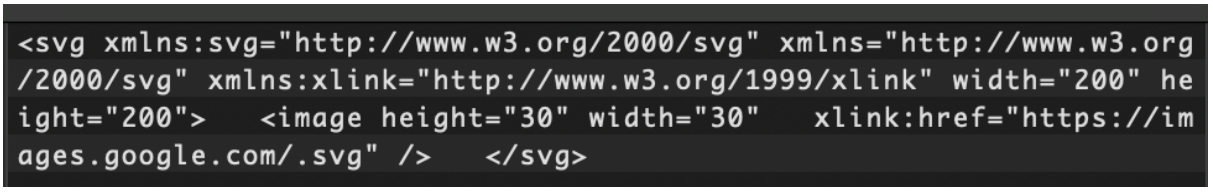
```
<html>      <!-- CSRF PoC - generated by Burp Suite Professional -->
      <body>      <script>history.pushState('', '', '/')</script>
     <form action="https://www.gambiaemarket.com/user" method="POST"
>          <input type="hidden" name="csrf&#95;test&#95;name" value=
"a47c8a94946e7a90e101dd8681b2dad1" />          <input type="hidden"
name="email" value="test&#64;gmail&#46;com" />          <input type=
"hidden" name="password" value="&#64;&#64;mM12345678" />          <i
nput type="hidden" name="login" value="" />          <input type="su
bmit" value="Submit request" />        </form>      </body>    </htm
l>
```

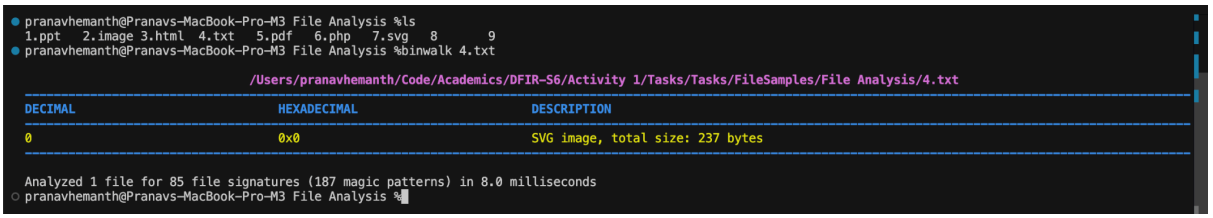
```
pranavhemanth@Pranavs-MacBook-Pro-M3 File Analysis %ls
1.ppt   2.image 3.html  4.txt   5.pdf   6.php   7.svg   8       9
pranavhemanth@Pranavs-MacBook-Pro-M3 File Analysis %binwalk 3.html
Analyzed 1 file for 85 file signatures (187 magic patterns) in 5.0 milliseconds
pranavhemanth@Pranavs-MacBook-Pro-M3 File Analysis %
```

This is a html file with multiple signatures

4.txt

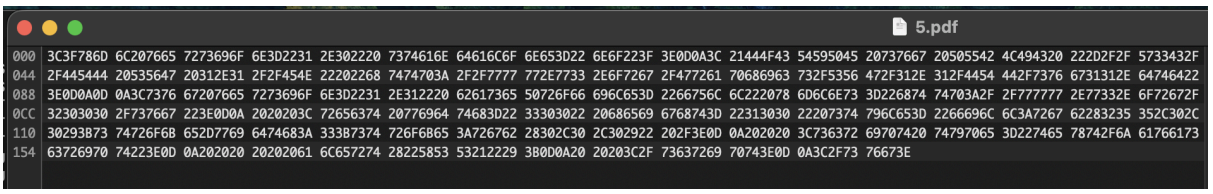






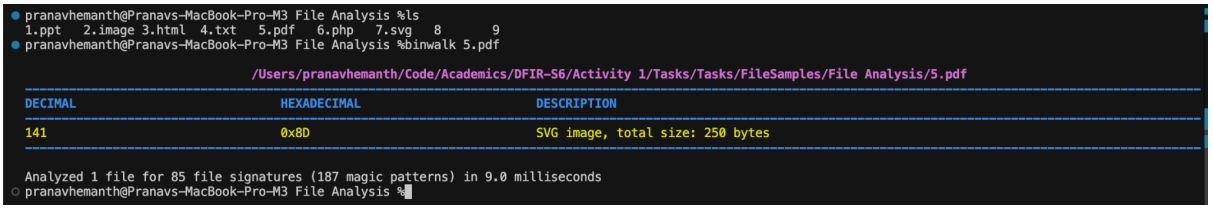
We can see that the file is an svg

5.pdf

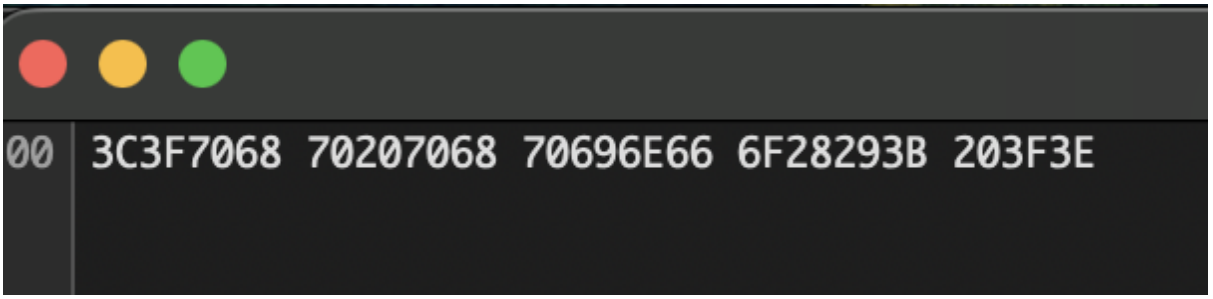




Again we can see that the file is an svg not a pdf as the extension says
We can confirm this with binwalk:



6.php



```
<?php phpinfo(); ?>
```

```
pranavhemanth@Pranavs-MacBook-Pro-M3 File Analysis %ls
1.ppt   2.image 3.html  4.txt   5.pdf   6.php   7.svg   8       9
pranavhemanth@Pranavs-MacBook-Pro-M3 File Analysis %binwalk 6.php
Analyzed 1 file for 85 file signatures (187 magic patterns) in 1.0 milliseconds
pranavhemanth@Pranavs-MacBook-Pro-M3 File Analysis %
```

Above is indeed a php file as the extension says

7.svg

7.svg

```
000 3C3F786D 6C207665 7273696F 6E3D2231 2E302220 7374616E 64616C6F 6E653D22 79657322 3F3E0D0A 3C21444F 43545950 4520666F 6F205B20 3C21454E 54495459 20666574
044 63682053 59535445 4D202266 696C653A 2F2F2F65 74632F70 61737377 64223E5D 3E0D0A3C 73766720 77696474 683D2231 32387078 22206865 69676874 3D223132 38707822
088 20786D6C 6E733D22 68747470 3A2F2F77 77772E77 332E6F72 672F3230 30302F73 7667220D 0A786D6C 6E733A78 6C696E6B 3D226874 74703A2F 2F777777 2E77332E 6F72672F
0CC 31393939 2F786C69 6E6B2220 76657273 696F6E3D 22312E31 223E0D0A 3C746578 7420666F 6E742D73 697A653D E2809C32 33222078 3DE2809C 38222079 3DE2809C 3238223E
110 26666574 63683B3C 2F746578 743E0D0A 3C2F7376 673E
```

```
<?xml version="1.0" standalone="yes"?>  <!DOCTYPE foo [ <!ENTITY fet
ch SYSTEM "file:///etc/passwd">]>  <svg width="128px" height="128px"
 xmlns="http://www.w3.org/2000/svg"  xmlns:xlink="http://www.w3.org/
1999/xlink" version="1.1">  <text font-size=...23" x=...8" y=...28">
&fetch;</text>  </svg>
```

```
pranavhemanth@Pranavs-MacBook-Pro-M3 File Analysis %ls
1.ppt   2.image 3.html  4.txt   5.pdf   6.php   7.svg   8       9
pranavhemanth@Pranavs-MacBook-Pro-M3 File Analysis %binwalk 7.svg

                /Users/pranavhemanth/Code/Academics/DFIR-S6/Activity 1/Tasks/Tasks/FileSamples/File Analysis/7.svg
DECIMAL                         HEXADECIMAL                     DESCRIPTION
103                             0x67                            SVG image, total size: 191 bytes

Analyzed 1 file for 85 file signatures (187 magic patterns) in 5.0 milliseconds
pranavhemanth@Pranavs-MacBook-Pro-M3 File Analysis %
```

It is indeed an svg file as we can see above

8

```
00 74657374 7374
```

```
testst
```

```
pranavhemanth@Pranavs-MacBook-Pro-M3 File Analysis %ls
1.ppt   2.image 3.html  4.txt   5.pdf   6.php   7.svg   8       9
pranavhemanth@Pranavs-MacBook-Pro-M3 File Analysis %binwalk 8
Analyzed 1 file for 85 file signatures (187 magic patterns) in 6.0 milliseconds
pranavhemanth@Pranavs-MacBook-Pro-M3 File Analysis %
```

It is a raw file as we can see above. It has no extension and is a simple raw text file

9

```
9
00 3C3F786D 6C207665 7273696F 6E3D2231 2E30223F 3E0D0A3C 21444F43 54595045 2061205B 3C21454E 54495459 20746573 74202254 48495320 49532041 20535452 494E4721
44 223E5D3E 0D0A3C6D 6574686F 6443616C 6C3E3C6D 6574686F 644E616D 653E2674 6573743B 3C2F6D65 74686F64 4E616D65 3E3C2F6D 6574686F 6443616C 6C3E
```

```
<?xml version="1.0"?>  <!DOCTYPE a [<!ENTITY test "THIS IS A STRING!
">]>  <methodCall><methodName>&test;</methodName></methodCall>
```

```
pranavhemanth@Pranavs-MacBook-Pro-M3 File Analysis %ls
1.ppt   2.image 3.html  4.txt   5.pdf   6.php   7.svg   8       9
pranavhemanth@Pranavs-MacBook-Pro-M3 File Analysis %binwalk 9
Analyzed 1 file for 85 file signatures (187 magic patterns) in 5.0 milliseconds
pranavhemanth@Pranavs-MacBook-Pro-M3 File Analysis %
```

The above is an xml file as it starts with 3C 3F 78 6D 6C. This is the header of an xml

22. Examine files of various formats given in ImageFiles Folder using File Viewer and understand if they need further investigation. Also check the File Properties option to understand more on file properties.

```
pranavhemanth@Pranavs-MacBook-Pro-M3 Image Files %ls
1(2).jpg
1200px-Olympic_rings_without_rims.svg.png
1200px-Peace_sign.svg.png
1200px-yin_yang-svg.png
1f5dc196213fbcd85801d35ac2dfcde5.gif
250px-Raelian_symbol.svg.png
3-2-cartoon-free-png-image-thumb.png
31154266_400x400.jpg
520px-Biohazard_symbol_(blue).mp4
8fb16268171047d65e4600e544ffcbd6.jpg
awen.gif
cartoon-article.jpg
cartoon-hd-poster-art-pncal19975-bpncal19975-large-original-imaehwt3p7fjhn7d.mp3
da2b622ecc9b84375187d770184bc450.png
ece437a85ccdb60e0d6d3adce3597e31.raw
Fan-oven.png
Flowers_123.jpg
Friends2.jpg
garry-norris-main-beach-sunrise.jpg
giphy_s.gif
Hashes.txt
images (1).jpg
images (1).png
images-of-cartoons-14.jpg
images.jpg
images.png
Kitty.jpg
maxresdefault.jpg
mip.jpg
Model.png
Pagan-Symbols-and-Their-Meanings-Pentagram.png
peacesign.jpg
resized_celtic-symbols-and-their-meanings-triskele.jpg
symbols-transparent-aboriginal.png
Tweety1.gif
u3Y7lq8z_400x400.jpeg
pranavhemanth@Pranavs-MacBook-Pro-M3 Image Files %
```

There are too many files to check but on manual check they all render in viewer (preview on macos) with issues.

We can try doing this more methodically by checking file type against extension:

```
pranavhemanth@Pranavs-MacBook-Pro-M3 Image Files %clear
pranavhemanth@Pranavs-MacBook-Pro-M3 Image Files %file *
1(2).jpg:                                                                       JPEG image data, JFIF standard 1.01, resolution (DPI), density 144x144, segment
 length 16, baseline, precision 8, 495x490, components 3
1200px-Olympic_rings_without_rims.svg.png:                                      PNG image data, 1200 x 554, 8-bit/color RGBA, non-interlaced
1200px-Peace_sign.svg.png:                                                      PNG image data, 1200 x 1200, 8-bit gray+alpha, non-interlaced
1200px-yin_yang-svg.png:                                                        PNG image data, 300 x 300, 8-bit gray+alpha, non-interlaced
1f5dc196213fbcd85801d35ac2dfcde5.gif:                                           GIF image data, version 89a, 3484 x 3484
250px-Raelian_symbol.svg.png:                                                   PNG image data, 250 x 277, 8-bit colormap, non-interlaced
3-2-cartoon-free-png-image-thumb.png:                                           PNG image data, 200 x 200, 8-bit/color RGBA, non-interlaced
31154266_400x400.jpg:                                                           JPEG image data, JFIF standard 1.01, resolution (DPI), density 96x96, segment l
ength 16, baseline, precision 8, 361x400, components 3
520px-Biohazard_symbol_(blue).mp4:                                              PNG image data, 520 x 520, 8-bit/color RGBA, non-interlaced
8fb16268171047d65e4600e544ffcbd6.jpg:                                           JPEG image data, JFIF standard 1.01, resolution (DPI), density 72x72, segment l
ength 16, progressive, precision 8, 502x790, components 3
awen.gif:                                                                       GIF image data, version 89a, 452 x 452
cartoon-article.jpg:                                                            JPEG image data, Exif standard: [TIFF image data, little-endian, direntries=0],
 baseline, precision 8, 621x330, components 3
cartoon-hd-poster-art-pncal19975-bpncal19975-large-original-imaehwt3p7fjhn7d.mp3: JPEG image data, JFIF standard 1.01, aspect ratio, density 1x1, segment length
16, baseline, precision 8, 588x704, components 3
da2b622ecc9b84375187d770184bc450.png:                                           JPEG image data, JFIF standard 1.01, resolution (DPI), density 72x72, segment l
ength 16, progressive, precision 8, 600x374, components 3
ece437a85ccdb60e0d6d3adce3597e31.raw:                                           JPEG image data, JFIF standard 1.01, resolution (DPI), density 72x72, segment l
ength 16, progressive, precision 8, 1000x985, components 3
Fan-oven.png:                                                                   PNG image data, 312 x 275, 8-bit colormap, non-interlaced
Flowers_123.jpg:                                                                JPEG image data, JFIF standard 1.01, resolution (DPI), density 72x72, segment l
ength 16, baseline, precision 8, 4896x3264, components 3
Friends2.jpg:                                                                   JPEG image data, Exif standard: [TIFF image data, big-endian, direntries=12, he
ight=469, bps=0, PhotometricIntepretation=RGB, orientation=upper-left, width=644], progressive, precision 8, 600x437, components 3
garry-norris-main-beach-sunrise.jpg:                                            JPEG image data, JFIF standard 1.01, aspect ratio, density 1x1, segment length
16, progressive, precision 8, 1048x699, components 3
giphy_s.gif:                                                                    GIF image data, version 89a, 350 x 200
Hashes.txt:                                                                     ASCII text, with CRLF line terminators
images (1).jpg:                                                                 JPEG image data, JFIF standard 1.01, aspect ratio, density 1x1, segment length
16, baseline, precision 8, 190x265, components 3
images (1).png:                                                                 PNG image data, 225 x 225, 8-bit colormap, non-interlaced
images-of-cartoons-14.jpg:                                                      JPEG image data, JFIF standard 1.01, aspect ratio, density 1x1, segment length
16, progressive, precision 8, 320x320, components 3
images.jpg:                                                                     JPEG image data, JFIF standard 1.01, aspect ratio, density 1x1, segment length
16, baseline, precision 8, 300x168, components 3
images.png:                                                                     PNG image data, 225 x 225, 8-bit colormap, non-interlaced
Kitty.jpg:                                                                      JPEG image data, JFIF standard 1.01, resolution (DPI), density 72x72, segment l
ength 16, Exif Standard: [TIFF image data, big-endian, direntries=12, height=456, bps=0, PhotometricIntepretation=RGB, orientation=upper-left, width=375], baseli
ne, precision 8, 575x700, components 3
maxresdefault.jpg:                                                              JPEG image data, JFIF standard 1.01, aspect ratio, density 1x1, segment length
16, baseline, precision 8, 1280x720, components 3
mip.jpg:                                                                        JPEG image data, JFIF standard 1.01, aspect ratio, density 1x1, segment length
16, Exif Standard: [TIFF image data, big-endian, direntries=1], baseline, precision 8, 506x512, components 3
Model.png:                                                                      PNG image data, 203 x 270, 8-bit/color RGB, non-interlaced
Pagan-Symbols-and-Their-Meanings-Pentagram.png:                                 PNG image data, 991 x 1005, 8-bit colormap, non-interlaced
peacesign.jpg:                                                                  JPEG image data, JFIF standard 1.01, aspect ratio, density 1x1, segment length
16, baseline, precision 8, 630x630, components 3
resized_celtic-symbols-and-their-meanings-triskele.jpg:                         JPEG image data, JFIF standard 1.01, aspect ratio, density 1x1, segment length
16, progressive, precision 8, 570x356, components 1
symbols-transparent-aboriginal.png:                                             PNG image data, 480 x 480, 8-bit/color RGBA, non-interlaced
Tweety1.gif:                                                                    GIF image data, version 89a, 524 x 591
u3Y7lq8z_400x400.jpeg:                                                          JPEG image data, JFIF standard 1.01, aspect ratio, density 1x1, segment length
16, baseline, precision 8, 400x400, components 3
pranavhemanth@Pranavs-MacBook-Pro-M3 Image Files %
```

We observe that the mp4 and mp3 files are actually image files according to this

```
pranavhemanth@Pranavs-MacBook-Pro-M3 Image Files %file 520px-Biohazard_symbol_\(blue\).mp4
520px-Biohazard_symbol_(blue).mp4: PNG image data, 520 x 520, 8-bit/color RGBA, non-interlaced
pranavhemanth@Pranavs-MacBook-Pro-M3 Image Files %file cartoon-hd-poster-art-pncal19975-bpncal19975-large-original-imaehwt3p7fjhn7d.mp3
cartoon-hd-poster-art-pncal19975-bpncal19975-large-original-imaehwt3p7fjhn7d.mp3: JPEG image data, JFIF standard 1.01, aspect ratio, density 1x1, segment length
16, baseline, precision 8, 588x704, components 3
pranavhemanth@Pranavs-MacBook-Pro-M3 Image Files %
```

one is png and another is jpeg

we can confirm with binwalk

```
pranavhemanth@Pranavs-MacBook-Pro-M3 Image Files %binwalk 520px-Biohazard_symbol_\(blue\).mp4

                        /Users/pranavhemanth/Code/Academics/DFIR-S6/Activity 1/Tasks/Tasks/Image Files/520px-Biohazard_symbol_(blue).mp4
--------------------------------------------------------------------------------------------------------------------------------------------------------------
DECIMAL                         HEXADECIMAL                     DESCRIPTION
--------------------------------------------------------------------------------------------------------------------------------------------------------------
0                               0x0                             PNG image, total size: 26767 bytes
--------------------------------------------------------------------------------------------------------------------------------------------------------------

Analyzed 1 file for 85 file signatures (187 magic patterns) in 8.0 milliseconds
pranavhemanth@Pranavs-MacBook-Pro-M3 Image Files %binwalk cartoon-hd-poster-art-pncal19975-bpncal19975-large-original-imaehwt3p7fjhn7d.mp3

  /Users/pranavhemanth/Code/Academics/DFIR-S6/Activity 1/Tasks/Tasks/Image Files/cartoon-hd-poster-art-pncal19975-bpncal19975-large-original-imaehwt3p7fjhn7d.mp3
--------------------------------------------------------------------------------------------------------------------------------------------------------------
DECIMAL                         HEXADECIMAL                     DESCRIPTION
--------------------------------------------------------------------------------------------------------------------------------------------------------------
0                               0x0                             JPEG image, total size: 33196 bytes
--------------------------------------------------------------------------------------------------------------------------------------------------------------

Analyzed 1 file for 85 file signatures (187 magic patterns) in 5.0 milliseconds
pranavhemanth@Pranavs-MacBook-Pro-M3 Image Files %
```

23. Implementation of software write blocker using Thumbscrew
Explanation:

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Implementation of a software write blocker using Thumbscrew means configuring a system so that any forensic evidence drive can be read but never written to, thereby preserving the integrity of the original data during examination.

Using Thumbscrew

Thumbscrew is a forensic driver designed to block all write operations at the operating system level. When it is enabled, it intercepts disk I/O requests and prevents any write commands from reaching the connected storage device. To use it, the investigator first installs Thumbscrew, enables write blocking, and then connects the suspect drive. Even if the operating system or an application attempts to create, delete, or modify files, Thumbscrew stops those operations. This ensures that timestamps, file contents, and file system structures on the evidence drive remain unchanged throughout the analysis.

Using macOS as a software write blocker

macOS does not support Thumbscrew directly, but the same write blocking effect can be achieved by mounting evidence drives or disk images in read only mode. When a suspect drive is connected, it is unmounted and then remounted with the read only flag. This tells the operating system to allow only read operations and reject any write attempts. Any effort to modify the drive will result in an error, confirming that the evidence is protected.

In both cases, whether using Thumbscrew on Windows or read only mounts on macOS, the goal is the same: to prevent any accidental or intentional modification of forensic evidence while still allowing investigators to examine and extract data safely.